How the tolerance models were built, trained and tested

A plain-language guide to notebook `05_tolerance_score_modelling`, sections 3–16: which data went where, what each model learned from, and how accuracy was measured

Results as of 21 October 2026 (final test run at 17:52, plan `9d090042fbe2`). Every number below is taken from the notebook's outputs and the saved files in `results/tolerance_model/`.

FIRST, THE MOST IMPORTANT POINT

Your original goal was a model that **predicts the tolerance score**, so that the prediction itself becomes the tolerance score. That model exists (§11–§13) and was tested on the validation genes in §14. But the model tested on the untouched genes in §16 is a different design: it **predicts the ClinVar label** (pathogenic or benign) of each variant, and the tolerance score is one of its *inputs*. Section 1 explains the two designs side by side. Section 9 lays out three ways to get back to your original goal.

The short answers to your questions

How were RSA and the structural annotations included in the training data? They were attached to every residue by a lookup on *(gene, residue number)* in §7–§8, giving one big table with one row per residue. A variant then copies the row of the residue it sits on (section 4, Figure 8). They were inputs to two models: the tree model that predicts tolerance (14g), and the variant model (14h, §16). They were deliberately *not* inputs to the sequence-only networks of your original design, which see only the protein sequence.

How were ESM's weights and scores used? In three ways (Figure 7). (1) **Scores**: log-probabilities from ESM-1v became four numbers per variant. (2) **Representations**: ESM-1v's internal description of each residue, 1,280 numbers, fed the networks that predict tolerance. (3) **Fine-tuned weights** (LoRA on ESM-2) are planned for Biowulf but haven't been run. ESM's own weights have not been changed in any model trained so far.

How were ClinVar clinical significance and gnomAD allele frequency used? The ClinVar **classification** is the *answer* the variant model learns, and the yardstick for every accuracy number. ClinVar variant **positions** (any classification) feed the ClinVar side of the tolerance score. gnomAD variant **positions** feed the gnomAD side. The gnomAD **allele count** is used only as a filter (a variant must be seen at least twice). The gnomAD **allele frequency** is **not used by any model** (Figure 2).

What features was the model actually trained on? The final §16 model used 42 numbers per variant in seven groups, listed one by one in section 7, plus "was this value missing?" flags. The models of your original design used only the sequence: amino-acid tokens, or ESM's 1,280-number representations.

Which accuracy metrics improved? For the variant model, the main metric is the **AUROC**: the chance that a random benign variant gets a more benign score than a random pathogenic one. On the untouched test genes it rose from **0.62 (ESM-1v alone) to 0.89**, and where ESM is unsure from **0.55 (chance) to 0.88**. For the tolerance-predicting models, the metric is the **within-gene Spearman correlation**. It rose from 0.02 (ESM alone) to 0.125 at best, against a maximum possible of 0.39 (section 9, Figure 29).

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How to read this guide

- **§ numbers** (§3, §14, …) are section numbers in the notebook; **14a–14i** and **16a–16h** are its subsections.
- Sections 1–2 give the big picture. Sections 3–4 explain the data, section 5 the accuracy metrics, sections 6–8 the models and their results, and section 9 what it all means. Words in **bold** are explained where they first appear and again in the glossary.
- Figures marked "from 14f", "from 16g" and so on are copied unchanged from the notebook's outputs. The others were drawn for this guide from the same saved results. All figures are high resolution: zoom in on any detail.
- Two designs run through the whole guide. In **Design A** (your original goal) the model *predicts* the tolerance score. In **Design B** the model predicts the ClinVar label and is *given* the tolerance score as evidence. Section 1 explains both.
- Two different collections of genes are used for checking models. The **validation genes** were used to compare and choose models (§14). The **test genes** were used once, at the end, for the final check (§16). Results on test genes are the most trustworthy.

The running example

Most diagrams use one gene, **GLRA1** (the α1 subunit of the glycine receptor, 449 residues), and one variant in it, **p.A300P**: alanine 300 replaced by proline, classified as pathogenic in ClinVar. GLRA1 is a training gene, so its data were used to fit the models but not to judge them.

1. Two ways to build a tolerance model

A **machine-learning model** is a formula with adjustable numbers (its **weights**). It is given **inputs** (also called **features**: everything it is allowed to look at) and produces an **output**. While it **trains**, its output is compared with the correct answer (the **target**), and the weights are adjusted to make the two agree better. After training, the model is tested on data it has never seen. The choice of target decides what the model becomes.

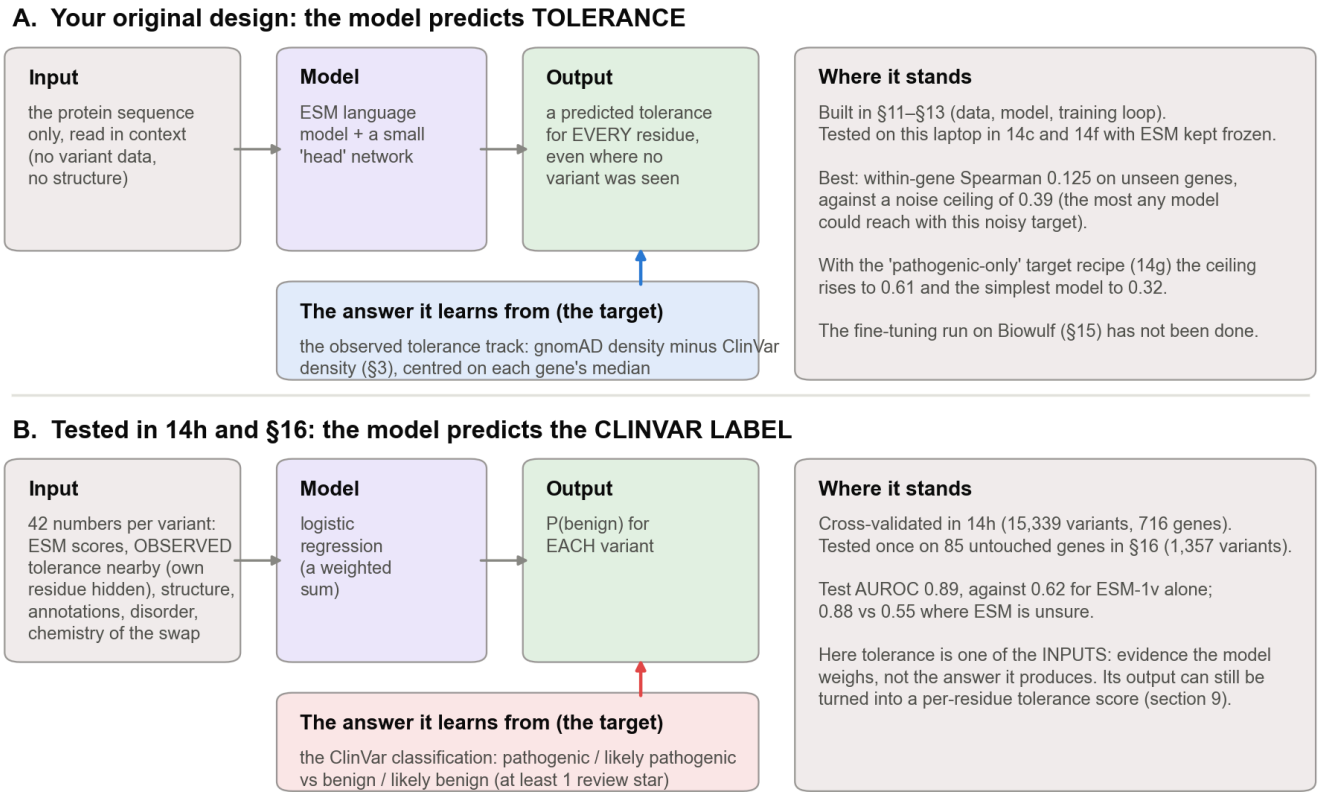


Figure 1. The two designs in this notebook. In Design A (your original goal) the tolerance score is the *target*: the model learns to predict it from the protein sequence. In Design B, the ClinVar label is the target, and the observed tolerance score is one of 42 *inputs*.

Design A: predict tolerance (the original goal)

The observed tolerance score (section 3) can only be computed where human variants have been seen, and it is noisy. A model that predicts it from the protein sequence alone could give a tolerance value for *every* residue of *any* protein, smoothing out the noise. §11 turned every protein into model input, §12 built the model (ESM-2 with a small "head" network), and §13 tested the code on this laptop. §14 then asked whether the target can be learned at all.

Design B: predict the clinical label (what §16 tested)

Your underlying hypothesis is that human population data mark constraint that evolution (ESM) misses. The most direct test of that hypothesis is at the level of single variants: put the human-variant evidence and the evolutionary evidence side by side, and see whether together they separate known pathogenic from known benign variants better than ESM alone.

Why the notebook moved from A to B in §14

- **The tolerance target is mostly noise at the level of single residues.** Built twice from two random halves of the variants, the two versions barely agree. Even a perfect model could reach a correlation of only about **0.39** with it (section 3).
- **The best Design A model reached 0.125**, about a third of that ceiling (section 6).
- **As a way to classify variants, predicted tolerance did worse than ESM alone** (AUROC 0.50–0.79 vs 0.94 on the validation genes), because one tolerance value per residue can't tell a mild swap from a drastic one at the same position.
- **Design B worked well**: combining tolerance with ESM lifted the AUROC well above ESM alone (section 7). So 14i recommended making it the main product, and §16 tested it.

IN PLAIN WORDS

Your original model predicts *how tolerant each residue is*. The tested model predicts *whether a specific variant is benign*, and uses the tolerance score as one of its clues. Design A has been built and tried, but not tested on the untouched genes, and its GPU fine-tuning hasn't been run. Section 9 shows how the two designs can be reconciled.

2. The ingredients, and where each one went

Everything in the notebook is built from the 1,082 postsynaptic SynGO genes, using one reference protein sequence per gene (the MANE transcript). The table lists every data source; Figure 2 shows which model used which.

Source	What it is	What was taken from it
gnomAD v4	genetic variants in about 800,000 people, with those known to have severe childhood disease excluded (a "healthy" reference population)	1,081,180 missense variants in these genes; **432,831 used**: those seen in at least 2 people, with an extra rule for X-linked genes (section 3)
ClinVar	variants reported by clinical laboratories, each with a classification (pathogenic, benign, uncertain = VUS) and a review status (stars)	206,892 missense variants (any classification) for the tolerance score. As answers for Design B: **15,339** pathogenic / likely pathogenic or benign / likely benign variants with ≥ 1 star in the training and validation genes, and **1,357** in the test genes
AIUPred	a predictor of intrinsically disordered (floppy, unstructured) regions	a disorder score and a binding-region score for every residue
ESM-1v	a protein language model with 650 million weights, trained on natural protein sequences	log-probabilities for all 20 amino acids at every residue (§4); internal representations (14c)
Structures	experimental structures (PDB) and AlphaFold predictions	solvent accessibility (RSA) for 61% of residues, AlphaFold confidence (pLDDT), binding interfaces from PDBe-KB (§7)
Annotations	InterPro, UniProt, MobiDB, ELM, PDBe-KB	domains, motifs, PTMs, binding and ligand sites, transmembrane segments, curated disorder, conservation (§8)

	Builds the tolerance score (§3)	Design A: networks predict tolerance	Design A: trees predict tolerance (14g)	Design B: variant model (14h, §16)	Labels and evaluation
Protein sequence (MANE transcript)	● track length	● input (tokens)	● amino acid, position	● chemistry of the swap	–
gnomAD variants: where they are	● gnomAD density	–	–	● counts nearby, tolerance	–
gnomAD allele count	● filter: seen ≥ 2×	–	–	–	–
gnomAD allele frequency	–	–	–	–	–
gnomAD zygosity (X / Y genes)	● filter: X-linked rule	–	–	–	–
ClinVar variants: where they are	● ClinVar density	–	–	● counts nearby, tolerance	–
ClinVar classification (P / B / VUS)	○ tested only (14g)	–	–	–	● THE LABEL (P vs B)
ClinVar review stars	–	–	–	–	● filter: ≥ 1 star
AIUPred disorder and binding	● chooses the smoothing	–	● 3 features	● 3 features	–
ESM-1v scores (log-probabilities)	–	–	● 3 scores	● 4 scores incl. LLR	● baseline to beat
ESM-1v representations (1,280 / residue)	–	● input (frozen)	● 64 summaries	–	–
ESM-2 weights, fine-tuned with LoRA	–	○ planned (Biowulf)	–	–	–
Structure: RSA, pLDDT, interfaces (§7)	–	–	● 7 features	● 6 features	–
Annotations + conservation (§8)	–	–	● 17 features	● 17 features	–

● used as information (an input, or to build the score)
● used only to decide which variants count
● the answer, or what accuracy is measured against
○ planned, or tried only as a test

'–' = not used there. Note: gnomAD allele FREQUENCY was added to the variant table but is not used by any model.

Figure 2. Where each piece of data was used. Columns are the places data can go: building the tolerance score, the models of Design A, the variant model of Design B, and the labels that accuracy is measured against. A dash means "not used there". Two things stand out: the ClinVar *classification* is used only as the answer (never as an input), and the gnomAD allele *frequency* is not used anywhere.

ALLELE FREQUENCY: NOT USED, AND SAFER THAT WAY

Allele frequency was added to the variant table (a side request) but was never chosen as a model input. That turns out to be the safer choice. ClinVar often classifies a variant as benign *because* it is common in gnomAD, so giving the model a variant's own allele frequency would partly hand it the answer, and a high score would be circular. The variant model instead uses gnomAD variants *near* the variant, with the variant's own residue hidden (section 7).

3. Step 1: turning variants into a tolerance score (§3)

The idea: positions where healthy people carry many missense variants are probably **tolerant** of change, and positions where disease reports cluster are probably **intolerant**. Figure 3 builds the score for one gene, GLRA1 (a glycine-receptor subunit), which is used as the running example throughout this guide.

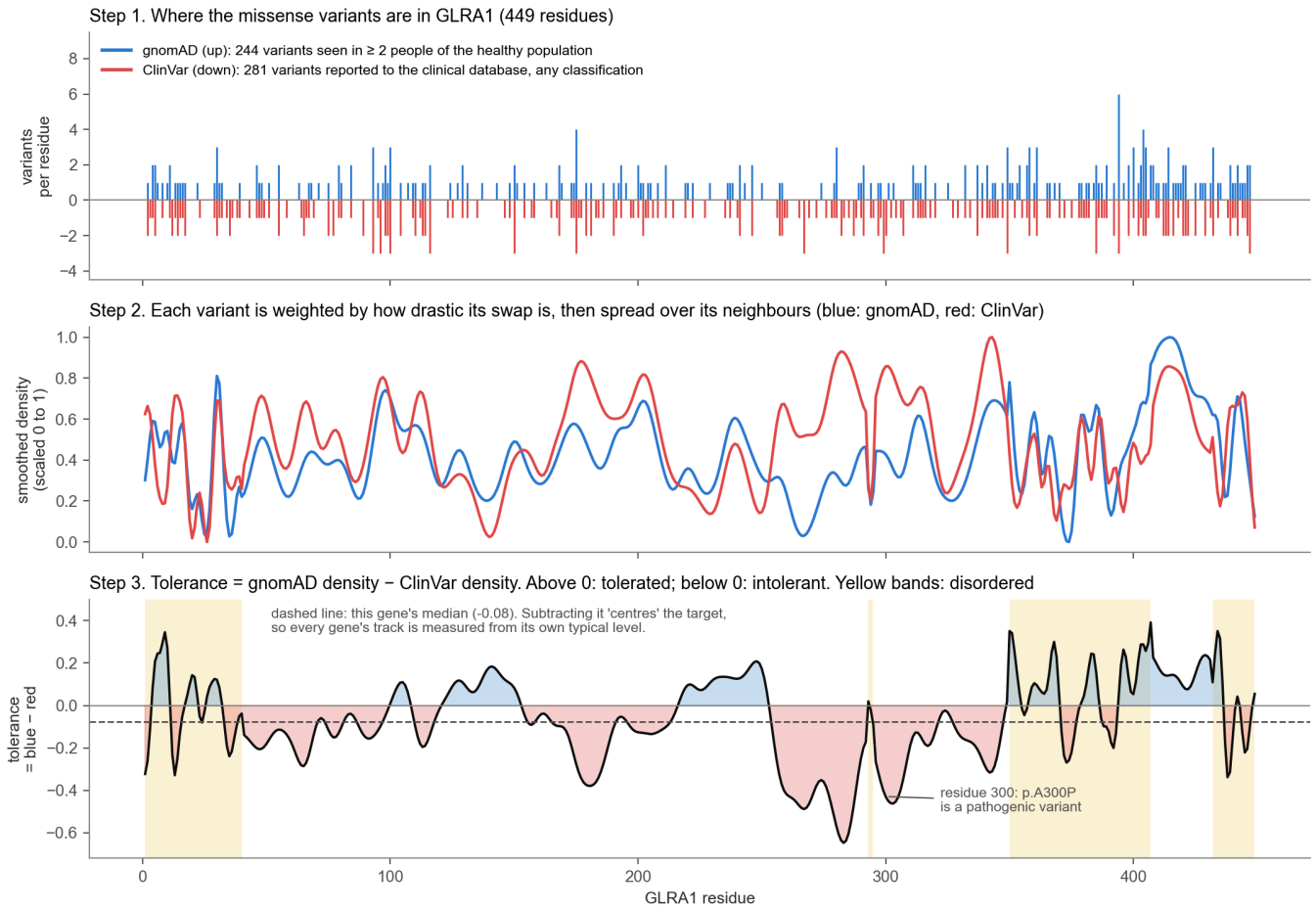


Figure 3. Building the tolerance score for GLRA1. Step 1: every gnomAD variant (up) and ClinVar variant (down) at its residue. Step 2: each set is turned into a smooth density. Step 3: tolerance = scaled gnomAD density − scaled ClinVar density. Red regions have more disease reports than healthy variation. Residue 300, where the pathogenic variant p.A300P sits, is in a deep intolerant region.

The steps in detail

1. **Choose which variants count.** gnomAD variants seen in at least 2 people (a variant seen once may be a sequencing error or one person's private variant). On the X and Y chromosomes, a gnomAD variant counts only if a man (with a single X) or a homozygous woman carries it. A variant seen only in women with a second, normal X could be a hidden recessive disease variant. All ClinVar missense variants count, *whatever their classification*.
2. **Weight each variant by how drastic it is** (Figure 4, right). Swapping alanine for serine is mild; tryptophan for cysteine is drastic. In ordered regions the BLOSUM62 table supplies this weight, in disordered regions the Grantham distance. If several variants sit at one residue, the most drastic one counts.
3. **Spread each variant over its neighbours** (Figure 4, left). Nearby residues usually share a fate, so each weight is spread with a bell-shaped curve: wide (±31 residues) in ordered regions, narrow (±5) in disordered

ones, where neighbours matter less. AIUPred's disorder score decides which applies.

4. **Subtract.** Each smoothed density is scaled to run from 0 to 1 within the gene, then tolerance = gnomAD − ClinVar, so it runs from −1 to +1.
5. **Centre.** Each gene's median is subtracted. This removes differences in overall level between genes (some genes simply have more ClinVar submissions), so the target describes which parts of a protein are more or less tolerant *than the rest of that protein*. This centred value, `tolerance_v4_centered`, is the target of Design A.

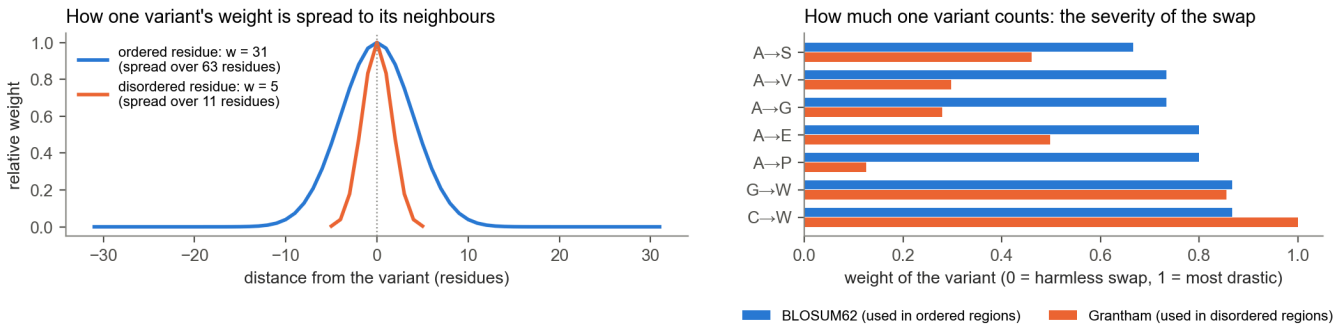


Figure 4. The two ingredients of the smoothing. Left: how far one variant's weight spreads. Right: how much one variant counts. BLOSUM62 weights differ little between swaps (0.67–0.87 here), so in ordered regions nearly every variant counts about the same. Grantham distances vary more. Grantham rates A→P as mild, even though a proline can break a helix: these tables measure chemistry, not structure.

How reliable is the tolerance score? The noise ceiling (14a)

To see how much of the score is signal and how much is chance, the variants were split randomly into two halves, and the score was built twice, once from each half. If the score were pure signal, the two versions would be nearly identical.

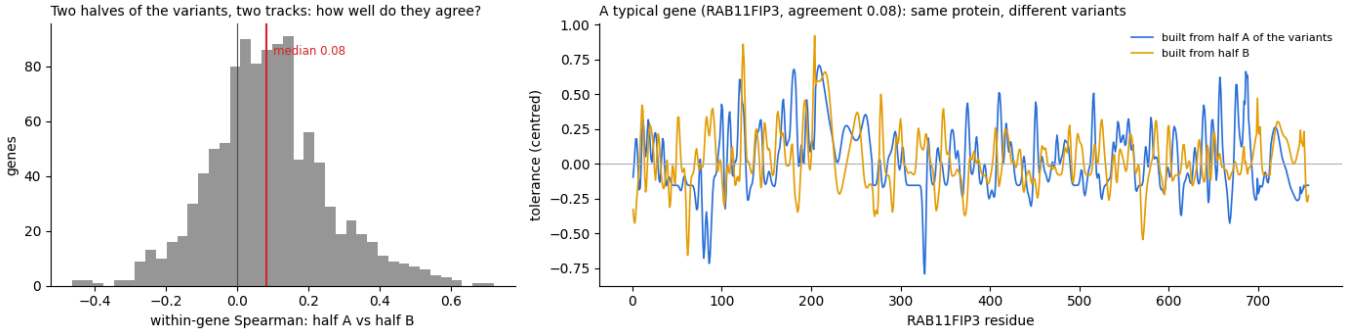


Figure 5. Two halves of the variants give two quite different tracks (from 14a). Left: the agreement (Spearman correlation) between the two half-tracks within each gene; the median is 0.08. Right: a typical gene.

From that agreement, the reliability of the full score is about 0.15, and the **noise ceiling** is √0.15 ≈ **0.39**: no model, however good, could correlate more than about 0.39 with this target, because the target itself is mostly noise at single-residue resolution. That is the yardstick for Design A (section 6). 14g later found a recipe with a much higher ceiling (0.61): building the ClinVar side from pathogenic variants only (section 6).

4. Step 2: evidence about each residue (§4, §7, §8)

ESM: what evolution says about each position

ESM is a protein language model. It was trained on tens of millions of natural protein sequences to fill in hidden amino acids, and in doing so learned which amino acids fit where. For each residue it gives a probability for every one of the 20 amino acids. Positions where evolution tolerates almost no change get a narrow, confident distribution; tolerant positions get a spread-out one.

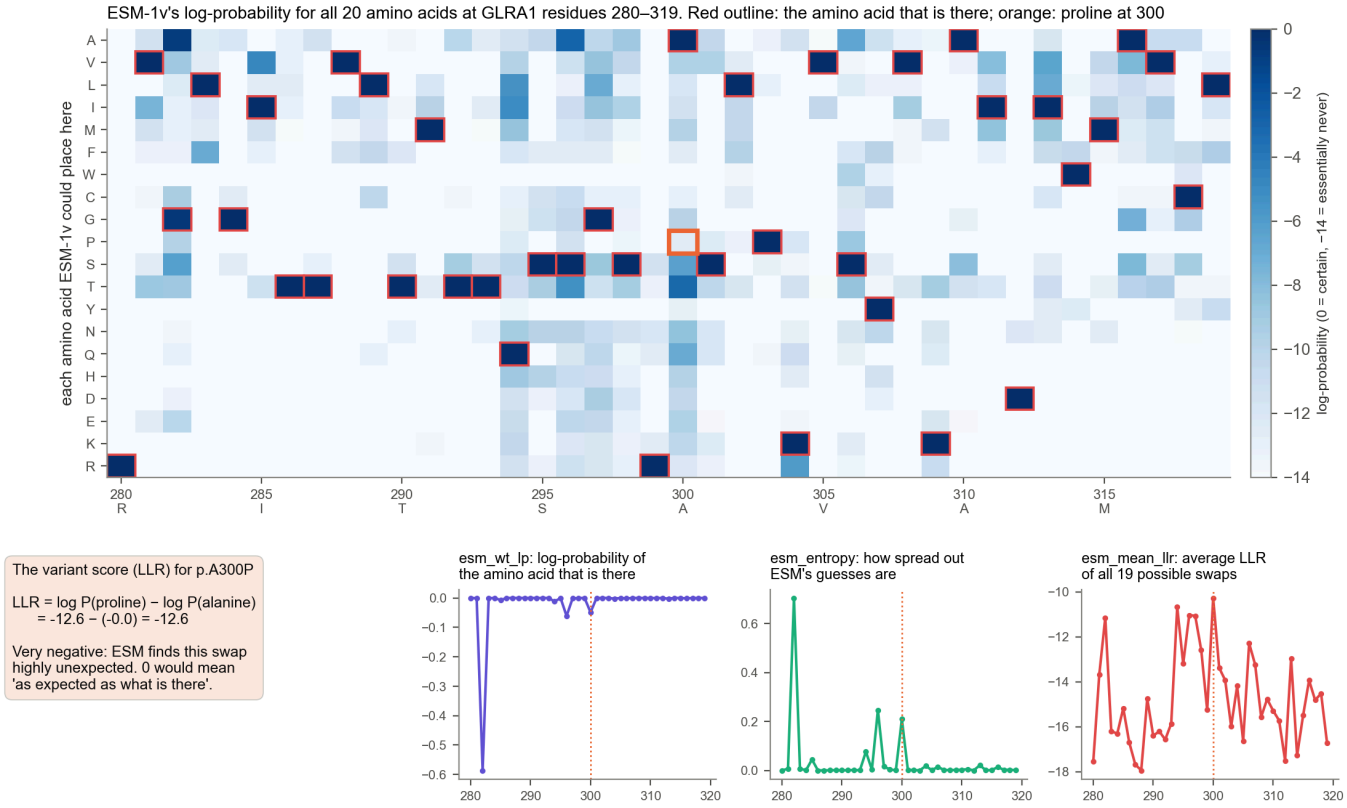


Figure 6. ESM-1v's view of 40 residues of GLRA1. Each column is a residue and each row an amino acid; darker means more likely. ESM-1v is shown the real sequence (the "wild-type marginal" method), so the amino acid that is there gets a probability close to 1. What carries information is how unlikely each *alternative* is. Bottom: the four ESM numbers used later. The **LLR** (log-likelihood ratio) of a substitution compares the new amino acid with the original: −12.6 for p.A300P means proline is far less likely than alanine here.

The four ESM numbers, all computed once in §4 and cached:

Name	Per	Meaning
`esm_llr`	variant	log P(new amino acid) − log P(original). Negative = the swap is unexpected; the more negative, the worse
`esm_wt_lp`	residue	log-probability of the amino acid that is there
`esm_entropy`	residue	how spread out ESM's probabilities are (low = a tightly constrained position)
`esm_mean_llr`	residue	the average LLR of all 19 possible swaps at this residue (one number per residue)

The three ways ESM was used

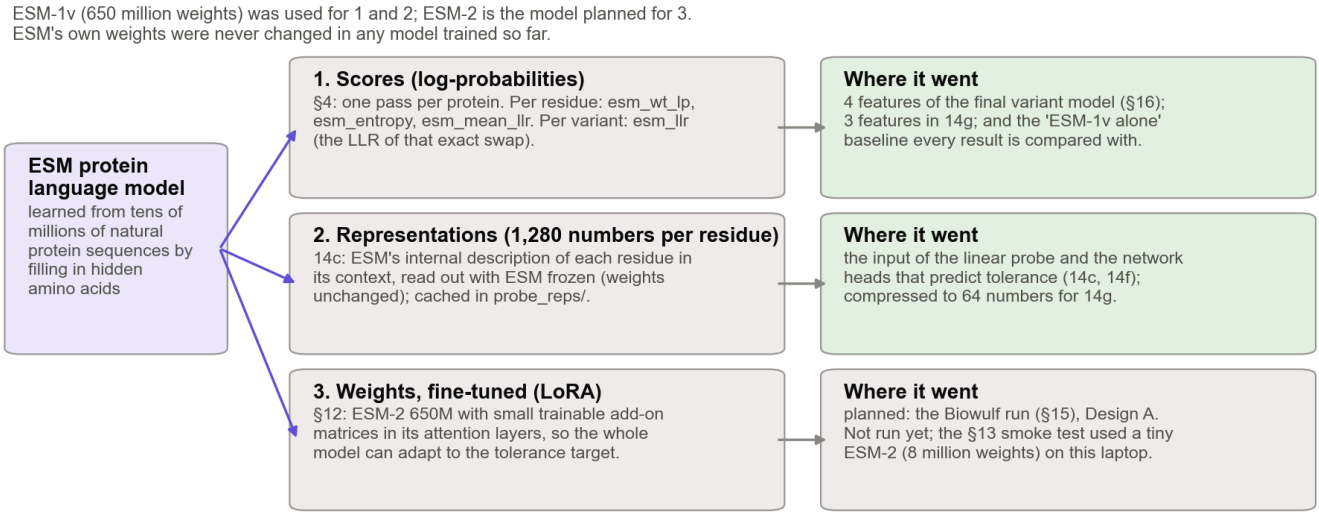


Figure 7. ESM's scores, representations and weights. Scores and representations come from ESM-1v with its weights frozen. Fine-tuning the weights themselves (way 3) was the plan for the Biowulf run of Design A.

A **representation** is ESM's internal description of a residue: a list of 1,280 numbers that summarises what ESM has inferred about that residue from the whole protein around it. It is far richer than the four scores, but it isn't readable directly. It becomes useful only when another model learns how to translate it into the quantity you care about, here tolerance.

Structure, annotations and disorder

- **RSA (relative solvent accessibility)**: the share of a residue's surface exposed to water, from 0 (buried in the core) to 1 (fully exposed). It was computed from the best experimental chain where one exists (PDB), otherwise from the AlphaFold model, and left empty where AlphaFold's confidence (**pLDDT**) is below 70. Available for 61% of residues; the rest are mostly disordered.
- **Interfaces**: whether the residue touches another chain in any experimental structure (from PDBe-KB): a copy of itself (`iface_homo`), a different protein (`iface_hetero`), or either (`iface_any`).
- **Annotations**: yes/no flags for domains, repeats, motifs, binding regions, linear interacting peptides, PTM sites, disulfides, ligand and active sites, conserved sites, coiled coils, transmembrane segments and signal peptides, plus a count of how many apply.
- **Disorder**: AIUPred's disorder and binding scores, and MobiDB's curated disorder. **Conservation**: MobiDB's PSI-BLAST conservation score, available for 53% of residues.

How it all comes together: one row per residue, one row per variant

Every source uses its own numbering (UniProt, PDB, AlphaFold), so §7 and §8 first translate everything onto the MANE sequence's numbering. Then every value is a column in one table, `residue_df`, with one row per residue: 780,750 rows in all. A variant needs only a lookup: *which gene, which residue?* This is how RSA and the annotations "entered the training data".

One row per residue (residue_df): every source lined up on the same numbering. GLRA1 residues 297–303

residue	amino acid	tolerance (centred)	ESM mean LLR	ESM entropy	RSA	pLDDT	interface	trans-membrane	disorder	conser-vation
297	G	-0.24	-11.1	0.02	0.50	92	yes	yes	0.40	2.5
298	S	-0.28	-12.6	0.00	0.07	90	yes	yes	0.38	1.2
299	R	-0.32	-15.3	0.00	0.35	90	yes	no	0.36	1.9
300	**A**	**-0.35**	**-10.3**	**0.21**	**0.64**	**90**	**yes**	**no**	**0.34**	**1.5**
301	S	-0.37	-13.4	0.00	0.38	84	yes	no	0.37	2.0
302	L	-0.38	-13.9	0.00	0.06	91	no	no	0.33	3.5
303	P	-0.39	-16.0	0.00	0.07	94	no	no	0.32	4.0

sequence | human variants (§3) | ESM-1v (§4) | structure (§7) | annotations / disorder / conservation (§8)

One row per variant: p.A300P (ClinVar: pathogenic, 1 review star)
= the whole residue-300 row above, looked up by (gene, position)
+ what depends on the exact swap: ESM LLR -12.6, BLOSUM62 and Grantham severity, change in hydrophobicity
+ the human-variant features recomputed with residue 300's own variants hidden: tolerance, counts within ±5 / ±15, gene-wide rates

Figure 8. From residue rows to a variant row. Top: seven rows of `residue_df` around GLRA1 residue 300, with columns coloured by source. Bottom: a variant's row is the row of its residue, plus what depends on the exact swap, plus the human-variant features recomputed with the variant's own residue hidden (section 7).

5. How accuracy is measured

Data the model has never seen

A model can look perfect on the data it learned from simply by memorising it (**overfitting**). Accuracy therefore always has to be measured on data kept aside. §10 split the genes three ways, keeping families of related genes (**clusters**: similar sequences or the same PANTHER family) together, so a test gene never has a close relative in training:

Split	Proteins	Residues	Used for
Training	854	618,540	learning the weights
Validation	107	79,309	comparing models and choosing settings (§14)
Test	107	78,366	one final, untouched check (§16)

Within the training genes, **cross-validation** gives every variant an honest score: the data are cut into 5 parts by cluster, the model is fitted on 4 parts and scores the 5th, and this repeats until every part has been scored (Figure 14, right).

For Design A: within-gene Spearman correlation

Design A predicts a whole track per protein. The question is whether the predicted track rises and falls where the real one does. The **Spearman correlation** compares the *ranks*: +1 means the residues are in exactly the same order, 0 means no relation, −1 means reversed. It is computed within each gene, and the median over the validation genes is reported. The noise ceiling (0.39) is the best possible score.

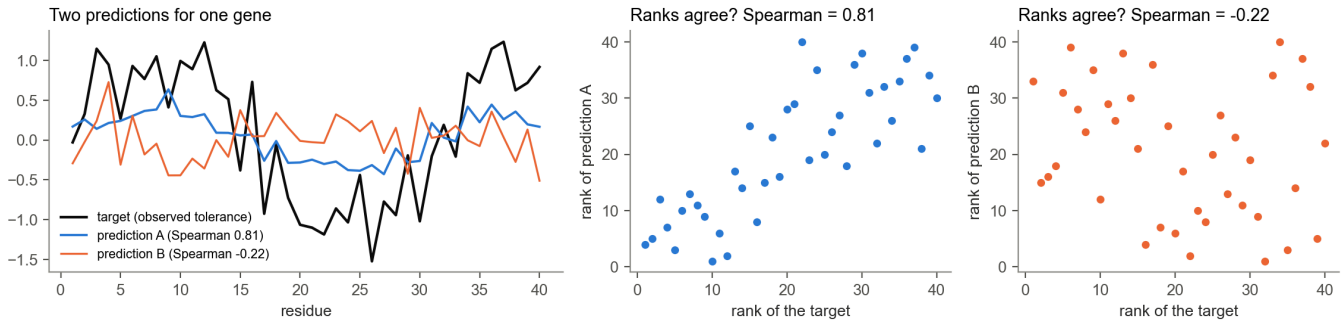


Figure 9. Spearman correlation, with made-up data. Prediction A follows the target's ups and downs (0.81) even though its values are smaller; prediction B doesn't (−0.22). Only the ranks matter.

For Design B: the AUROC

Design B gives each variant a score. Take every pair of one truly benign and one truly pathogenic variant, and count how often the benign one gets the more benign score. That share is the **AUROC** (area under the ROC curve). 0.5 is coin-flipping; 1.0 is perfect. It doesn't depend on choosing a cut-off, and it is the same as the area under the curve traced out by sliding a cut-off from one end of the scores to the other.



Figure 10. The AUROC, with 12 made-up variants. 31 of the 36 benign–pathogenic pairs are in the right order, so AUROC = 31/36 = 0.86. Right: the ROC curve; its area is the same 0.86.

Three versions of the AUROC

- **All variants**: the standard number, pooled over genes.
- **Within genes**: the average AUROC *inside* each gene with at least 3 pathogenic and 3 benign variants. This is the clinical question: in a gene we know, is *this* variant worse than *that* one? Pooling can hide or exaggerate differences (Figure 11), so both are reported.
- **Where ESM is unsure**: only the variants ESM-1v calls mild (its LLR is above the median for benign training variants). This is where human data were expected to help most.

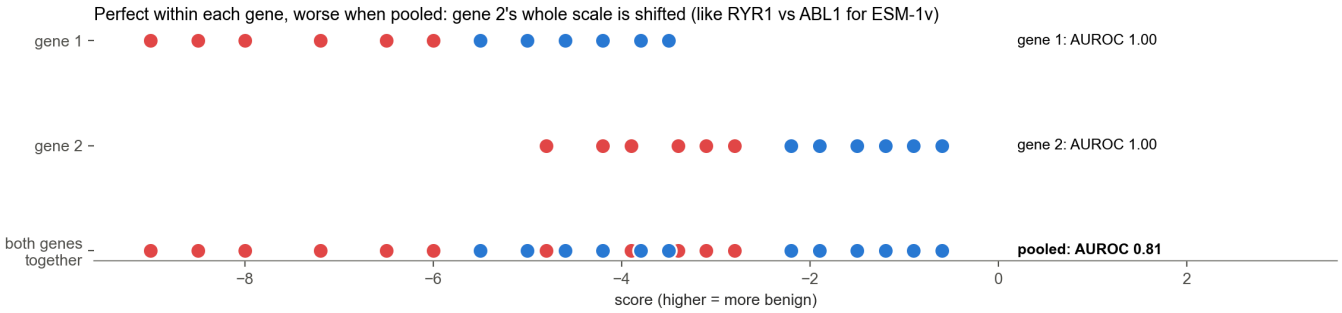


Figure 11. Why pooled and within-gene AUROC can disagree. Within each gene the score separates the classes perfectly, but gene 2's whole scale is shifted, so pooled across genes the AUROC drops to 0.81. This happened to ESM-1v on the test genes (section 8).

How sure are we? Bootstrap ranges

The test genes are just one sample of genes; a different sample would give somewhat different numbers. The **bootstrap** estimates how much: draw the test genes again at random, with replacement, 1,000 times, recompute the score each time, and keep the middle 95% of the results as the **95% range**. Whole genes are resampled, because variants in the same gene aren't independent. To compare two models, the *difference* is recomputed on each resample (a **paired** bootstrap). A comparison "passes" when the whole range is above 0.

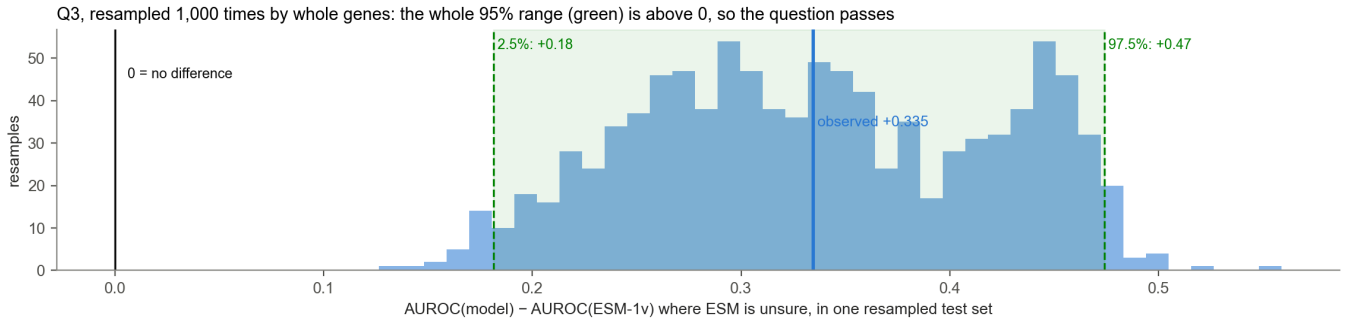


Figure 12. The bootstrap for one real result (§16, question Q3). The model's gain over ESM-1v, where ESM is unsure, was +0.335. Across 1,000 resamples of the test genes it stayed between +0.18 and +0.47, nowhere near 0, so the gain isn't a fluke of which genes landed in the test set.

Decisions and honest probabilities

The AUROC measures *ranking*. To make a call, a cut-off is needed: for example, call a variant benign when P(benign) ≥ 0.5. Then the useful numbers are the share of pathogenic variants caught, the share of benign variants correctly cleared, and the overall **accuracy**. **Calibration** asks whether a stated probability can be taken at face value: of the variants given P(benign) ≈ 0.8, are about 80% really benign? The **Brier score** (mean squared error of the probabilities; 0 is perfect) summarises it. Both are shown in section 8.

6. Design A in practice: predicting tolerance (14c, 14f, 14g)

Every Design A model gets the same target (the centred tolerance of each residue in the training genes) and is scored on the validation genes by within-gene Spearman correlation. They differ in what they see.

Model	Input (features)	How it learns	Section
Linear probe	ESM-1v's 1,280-number representation of each residue, ESM frozen	a weighted sum of the 1,280 numbers (ridge regression)	14c
From-scratch networks: window MLP, dilated CNN, BiLSTM	the amino-acid sequence only (tokens), up to 512 residues at a time	neural networks trained from random starting weights	14f
ESM heads: MLP, CNN, BiLSTM	ESM-1v representations, ESM frozen	small networks on top of ESM	14f
Gradient-boosted trees	97 numbers per residue: 64 ESM summaries, ESM scores, amino acid, disorder, conservation, structure, annotations, position	hundreds of small decision trees	14g
ESM-2 + LoRA (planned)	the sequence, through ESM-2 with trainable add-ons	fine-tuning on a GPU	§12, §15

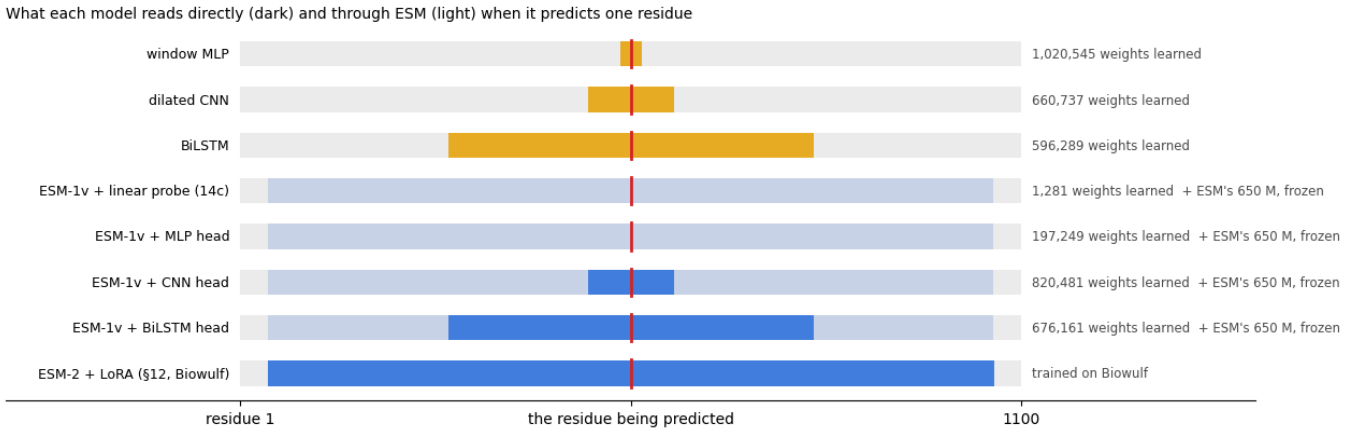


Figure 13. What each network can see when it predicts one residue (red) (from 14f). Dark: what it reads directly. Light: context that reaches it only through ESM, which read the whole protein. The right-hand column is the number of weights learned from our data.

How the networks were trained

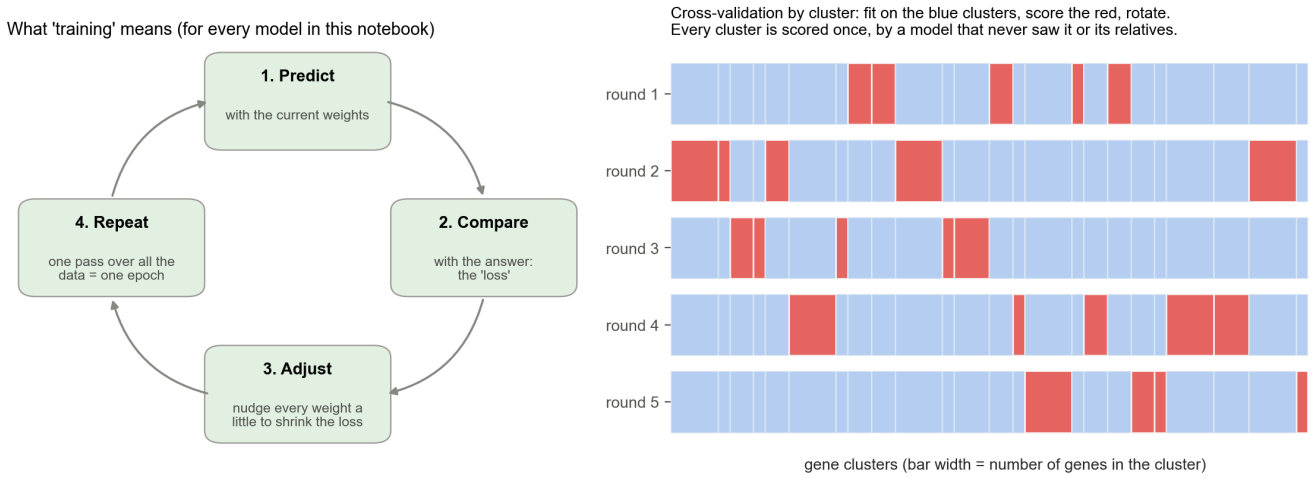


Figure 14. Training and cross-validation. Left: the loop every model repeats. Right: cross-validation by cluster, used in 14h.

- **Loss**: the Huber loss, a measure of how far the predicted track is from the target that is forgiving of the occasional large miss. Padding (empty space after short proteins) never counts.
- **Epochs**: each training protein contributes one random window of up to 512 residues per pass; 16 windows are processed at a time.
- **Early stopping**: 10% of the training clusters are set aside. After every epoch the model predicts them, and training stops when that score stops improving, keeping the best epoch. That is the point where the model starts to memorise rather than learn (**overfitting**).
- **Seeds**: each network was trained 3 times from different random starting weights, because small networks vary from run to run.

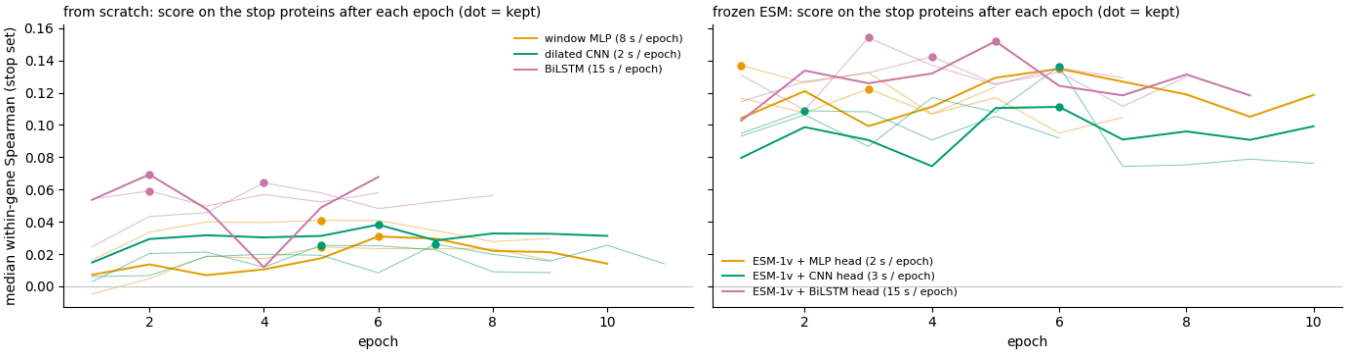


Figure 15. Early stopping in practice (from 14f). Each line is one model's score on the set-aside proteins after each epoch; the dot marks the epoch that was kept. Most models peak within a few epochs, a sign of how little learnable signal the target holds.

Results

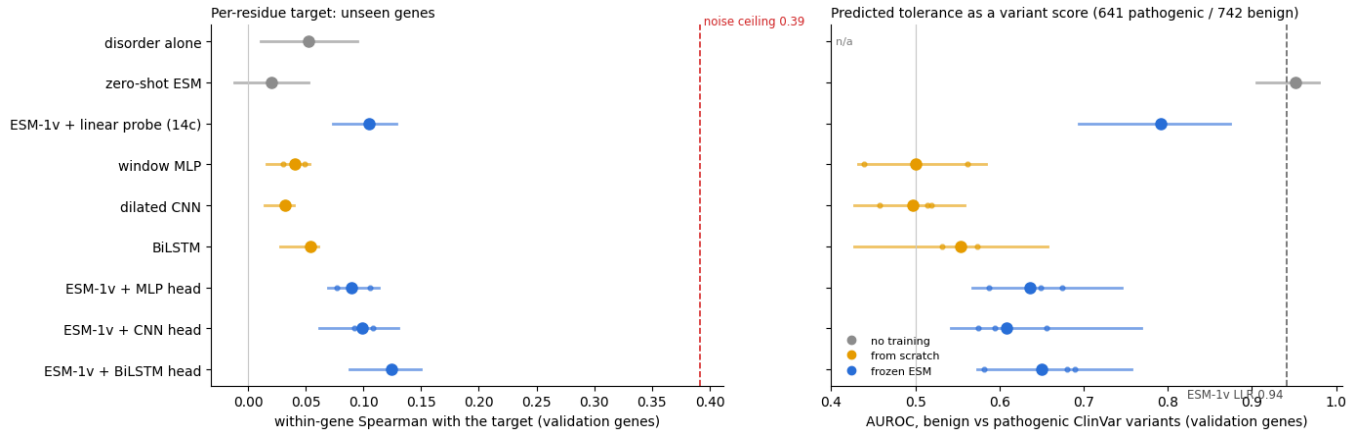


Figure 16. Design A results on the validation genes (from 14f). Left: within-gene Spearman with the target, against the noise ceiling (red dashes). Right: the predicted tolerance used to separate benign from pathogenic variants, against ESM-1v's own LLR (dashed line). Small dots are individual training runs; bars are 95% bootstrap ranges.

- **ESM's pre-training matters**: networks trained from scratch on ~850 proteins reached 0.03–0.05; the same networks on frozen ESM reached 0.09–0.125.
- **Context helps a little**: the BiLSTM, which reads the whole window in both directions, was the best of each group.
- **The best model reached 0.125, a third of the 0.39 ceiling.**
- **Predicted tolerance is a weak variant classifier** (AUROC 0.50–0.79), well below ESM-1v's LLR (0.94): one value per residue can't distinguish a mild swap from a drastic one at the same position.

Adding structure and annotations to predict tolerance (14g, part 1)

This is where RSA, interfaces and annotations were tried as inputs to Design A. A tree model got all 97 per-residue numbers, and each group was tested alone and removed in turn.

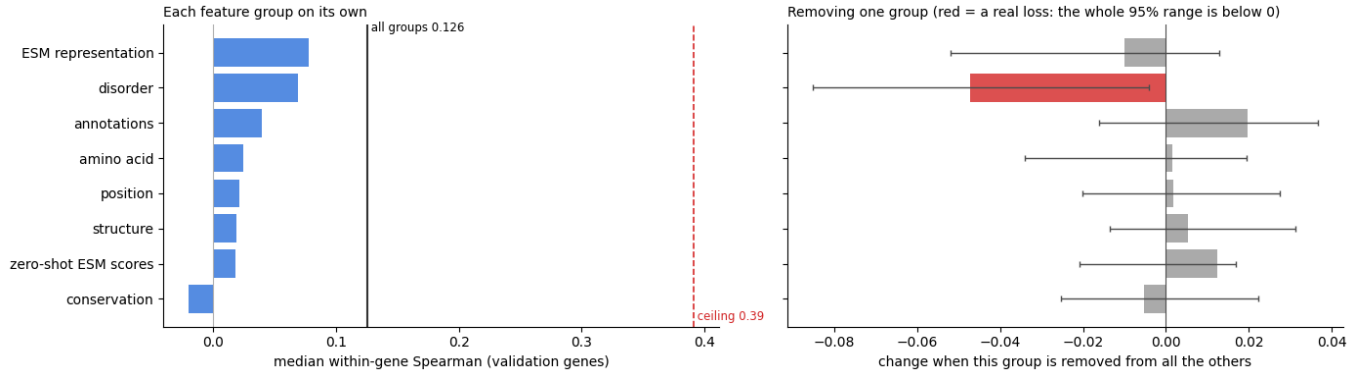


Figure 17. Which inputs help predict tolerance (from 14g). Left: each group alone. Right: the change when a group is removed from all the others, with 95% ranges. Only disorder is a real loss (red); every other change is within the noise.

All groups together reached 0.126, essentially the same as the best ESM head. Structure and annotations add nothing reliable to this target. One likely reason is that the target is so noisy that small, real effects disappear.

A better target (14g, part 2)

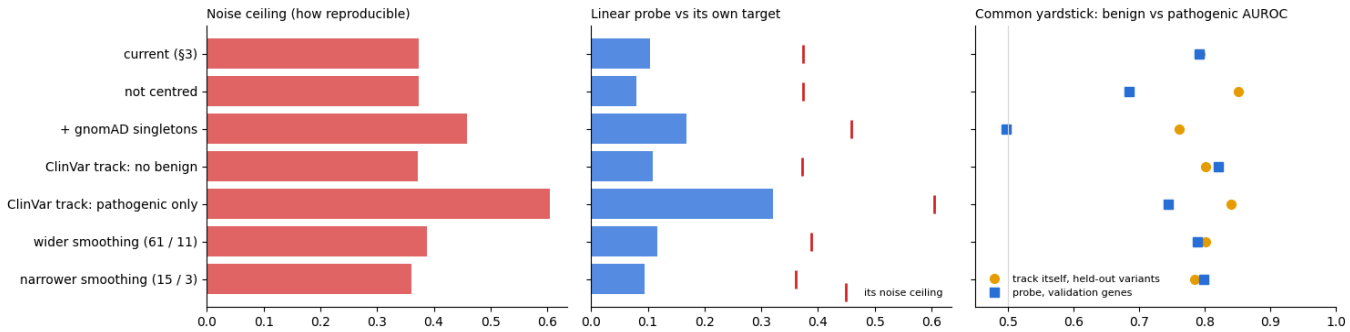


Figure 18. Changing the recipe of the target (from 14g). Left: how reproducible each recipe is (noise ceiling). Middle: the linear probe against its own target. Right: the common yardstick, the AUROC on held-out ClinVar variants.

Building the ClinVar side from **pathogenic variants only** (instead of every ClinVar entry, VUS and benign included) raised the ceiling from 0.37 to **0.61**, the probe from 0.10 to **0.32**, and the held-out AUROC of the track itself from 0.79 to **0.84**. Keeping gnomAD variants seen only once made the target more reproducible but less useful. This is the most promising lead for your original design (section 9).

7. Design B in practice: predicting the ClinVar label (14b, 14h)

The motivating result (14b)

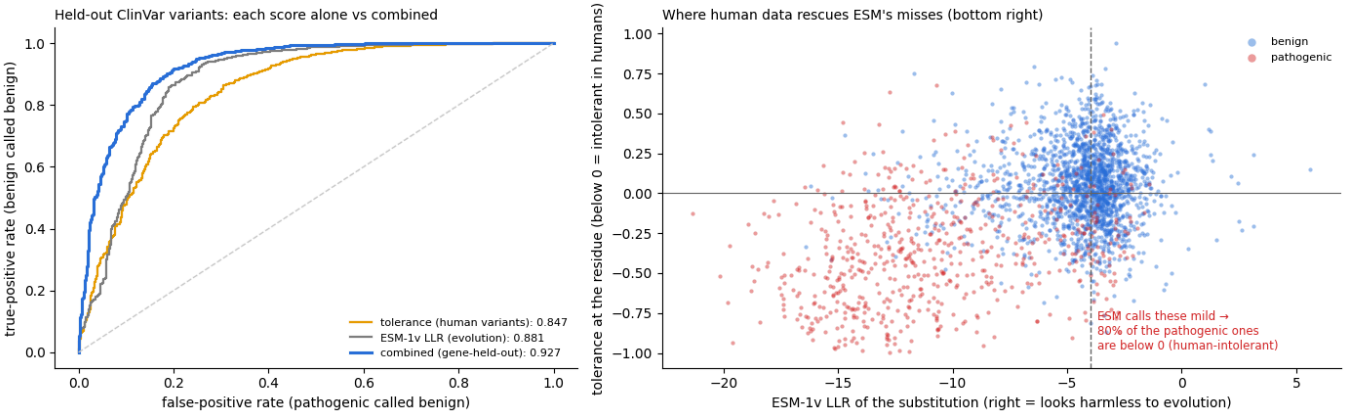


Figure 19. Human data and evolution are complementary (from 14b, on 2,477 held-out labelled variants). Left: tolerance alone (AUROC 0.847) and ESM-1v alone (0.881) both separate pathogenic from benign; combined, 0.927. Right: among pathogenic variants that ESM calls mild (bottom right), 80% sit in human-intolerant regions (tolerance below 0), against 42% of the benign ones.

The data and the answer

One row per labelled ClinVar variant: pathogenic / likely pathogenic versus benign / likely benign, with at least one review star. Conflicting and uncertain classifications were left out. The training and validation genes hold 15,339 such variants (5,449 pathogenic, 9,890 benign) in 716 genes; the test genes hold 1,357.

The 42 features the final model was trained on

Feature	What it measures
ESM scores (4) — evolution, from ESM-1v (§4)	
`esm_llr`	how unexpected this exact swap is: log P(new) − log P(original)
`esm_wt_lp`, `esm_entropy`, `esm_mean_llr`	how constrained the residue is: the original amino acid's log-probability, the spread of ESM's guesses, the average LLR of all 19 swaps
Human variants nearby (7) — from gnomAD and ClinVar, with the variant's own residue hidden	
`tolerance`	the §3 tolerance score at this residue
`gnomad_density`, `clinvar_density`	the two smoothed densities that make up the tolerance score
`gnomad_within_5`, `gnomad_within_15`	number of gnomAD variants within ±5 and ±15 residues
`clinvar_within_5`, `clinvar_within_15`	number of ClinVar variants (any classification) within ±5 and ±15 residues
Gene-wide variant rates (2)	

`gnomad_per_residue`, `clinvar_per_residue`	gnomAD and ClinVar variants per residue across the whole gene
Substitution type (3) — the chemistry of the swap	
`blosum62`, `grantham`	how different the two amino acids are (each scaled 0 to 1)
`hydrophobicity_change`	Kyte–Doolittle hydrophobicity of the new amino acid minus the original
Structure (6) — from §7	
`rsa`, `plddt`	how exposed the residue is; AlphaFold's confidence
`buried_sasa`	surface buried by partner molecules; computed for only a few showcase genes, so almost always empty
`iface_any`, `iface_homo`, `iface_hetero`	whether the residue touches another chain (any, a copy of itself, a different protein)
Annotations (16) — from §8	
`ann_*` (15 flags), `n_annotations`	yes/no for domain, repeat, motif (curated, predicted), binding region, linear interacting peptide (curated, predicted), PTM, disulfide, ligand site, active site, conserved site, coiled coil, transmembrane, signal peptide; and how many apply
Disorder and conservation (4) — AIUPred and MobiDB	
`disorder`, `binding`	AIUPred disorder score; AIUPred binding-region score
`ann_disorder_curated`, `msa_conservation`	curated disorder (yes/no); conservation across related sequences

Missing values (for example, no structure for a residue) are filled with the feature's median, and a "was it missing?" flag is added so the model can learn whether missingness matters. Every feature is then standardised (shifted and scaled to mean 0, spread 1), so the weights are comparable.

Hiding each variant's own residue

If a variant's own position fed its human-variant features, a pathogenic variant would partly vouch for itself, and so would a second pathogenic substitution at the same residue. So the residues with labelled variants are dealt into 5 groups, and the features of a variant in group *k* come from tracks rebuilt *without any variant at group-k residues*.

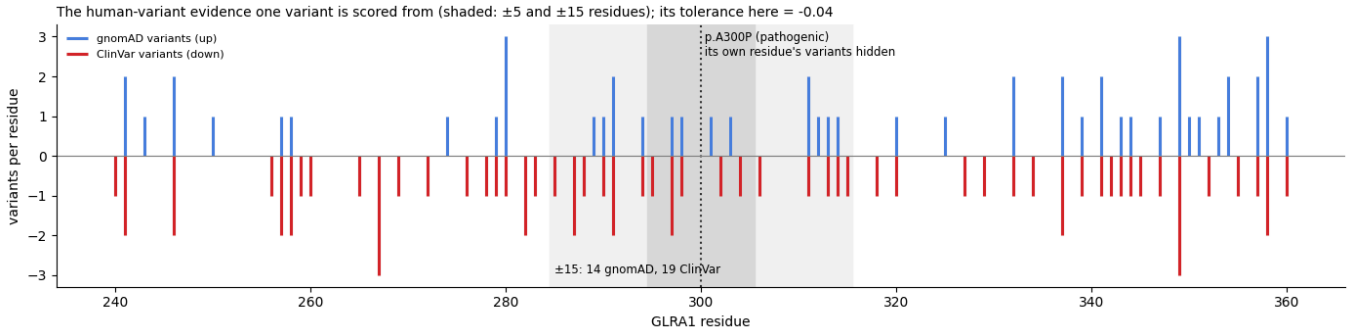


Figure 20. The evidence one variant is scored from (from 14h). The variant's own residue is hidden; the shaded windows show the ±5 and ±15 residues counted.

The model: logistic regression

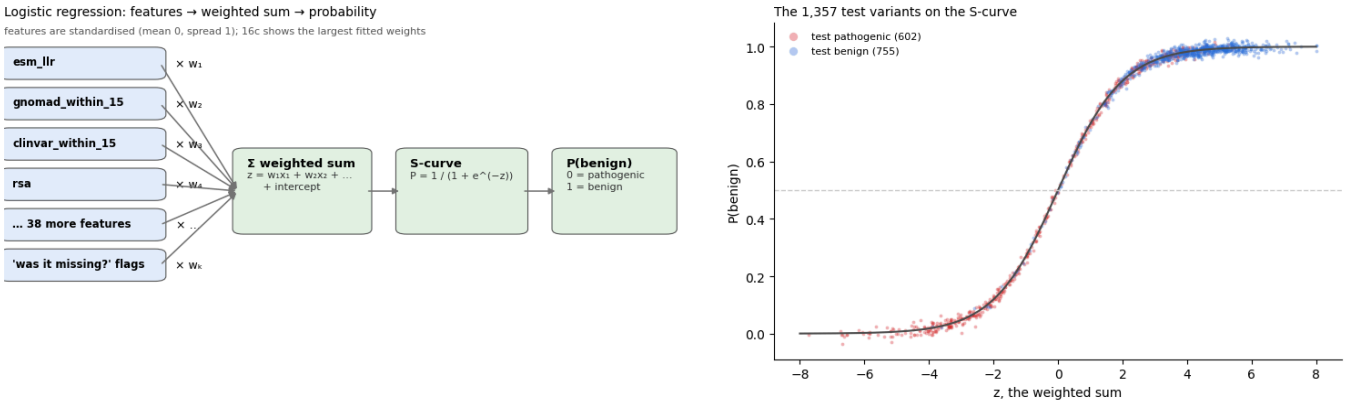


Figure 21. Logistic regression in three steps (from 16g). Each standardised feature is multiplied by its weight, the products are added up into one number *z*, and an S-shaped curve turns *z* into a probability of being benign. Right: the 1,357 test variants on that curve.

Training means choosing the 42 weights (plus those for the missing-value flags) that make the training labels as likely as possible, with a small penalty on very large weights. It takes a fraction of a second. Gradient-boosted trees, a small neural network (MLP) and two networks that read raw variant positions (window CNN and BiLSTM) were also tried in 14h, all with the same cross-validation by cluster (Figure 14, right).

Results of the cross-validation (14h)

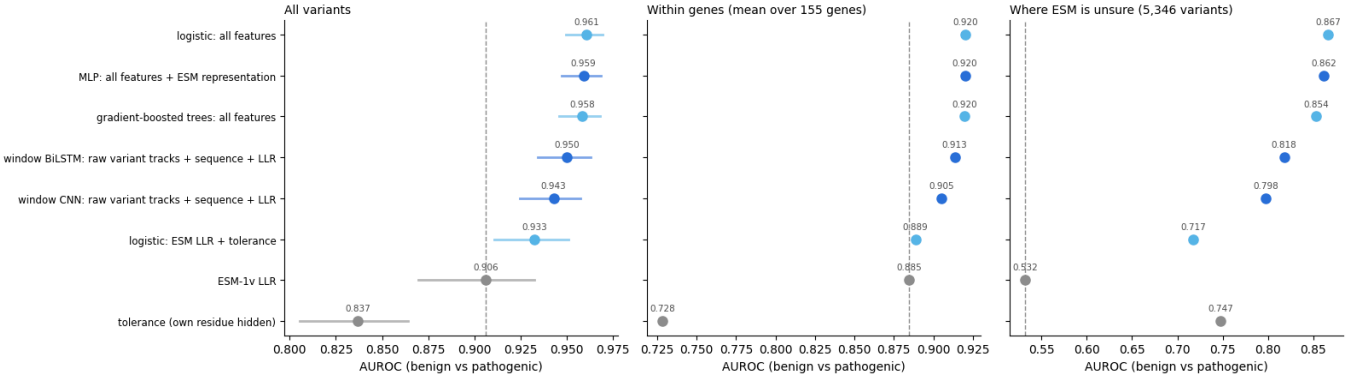


Figure 22. Every variant model, cross-validated on the training and validation genes (from 14h). The dashed line is ESM-1v alone. The same numbers are in the table below.

Model (14h cross-validation, 15,339 variants)	All variants	Within genes (155)	Where ESM is unsure
ESM-1v LLR alone (no training)	0.906	0.885	0.532
Tolerance alone, own residue hidden (no training)	0.837	0.728	0.747
Logistic regression: ESM LLR + tolerance	0.933	0.889	0.717
Logistic regression: all 42 features (chosen for §16)	**0.961**	**0.920**	**0.867**
Gradient-boosted trees: all features	0.958	0.920	0.854
MLP: all features + ESM representation	0.959	0.920	0.862
Window CNN: raw variant positions + sequence + LLR	0.943	0.905	0.798
Window BiLSTM: raw variant positions + sequence + LLR	0.950	0.913	0.818

- Combining all features lifts the AUROC from **0.906 (ESM-1v alone) to 0.961**; within genes from 0.885 to 0.920; where ESM is unsure from 0.53 (chance) to 0.87.
- Logistic regression, trees and the MLP all land at 0.958–0.961. The simplest one was chosen for the final test.
- The window networks, which learn their own smoothing from raw variant positions, came close (0.943–0.950) but didn't beat `convolve_v4` plus the other features.

Which feature groups matter?

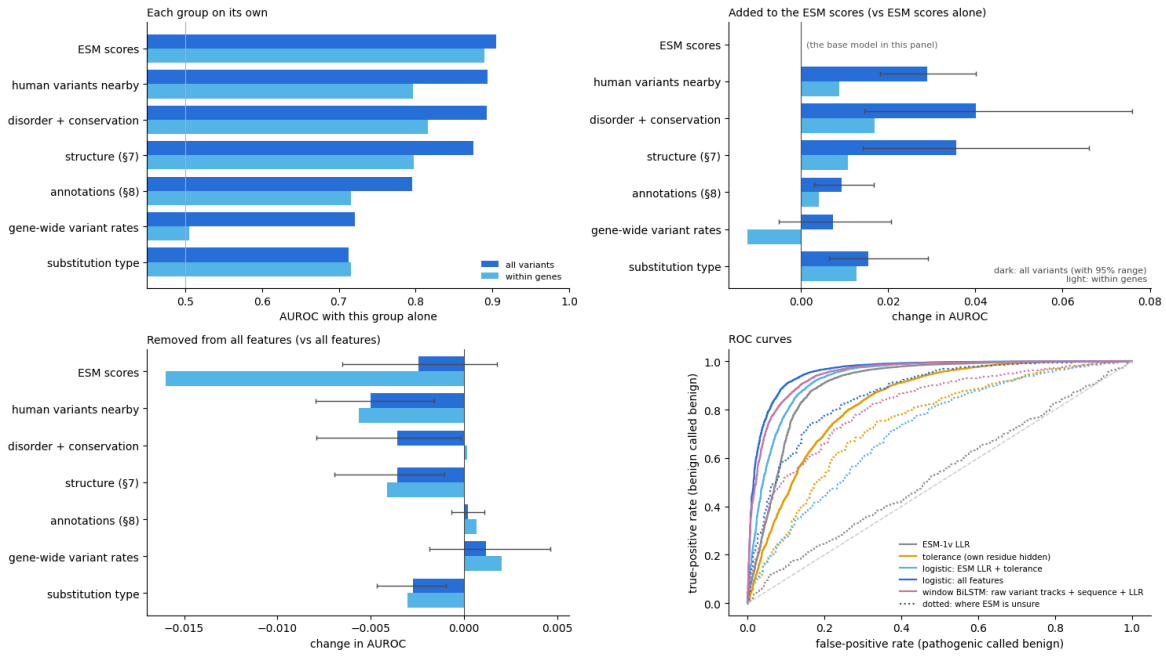


Figure 23. Feature groups, alone and in combination (from 14h, tree model). Top right is the direct test of the hypothesis: the gain when each group is added to the ESM scores.

Added to ESM, human variants nearby gain +0.029 AUROC, structure +0.036, disorder and conservation +0.040. All three are real gains, but removing any single group from the full model costs at most 0.005 on all variants: the groups largely carry the same information, because they all mark constrained regions of the protein.

8. The final test on untouched genes (§16)

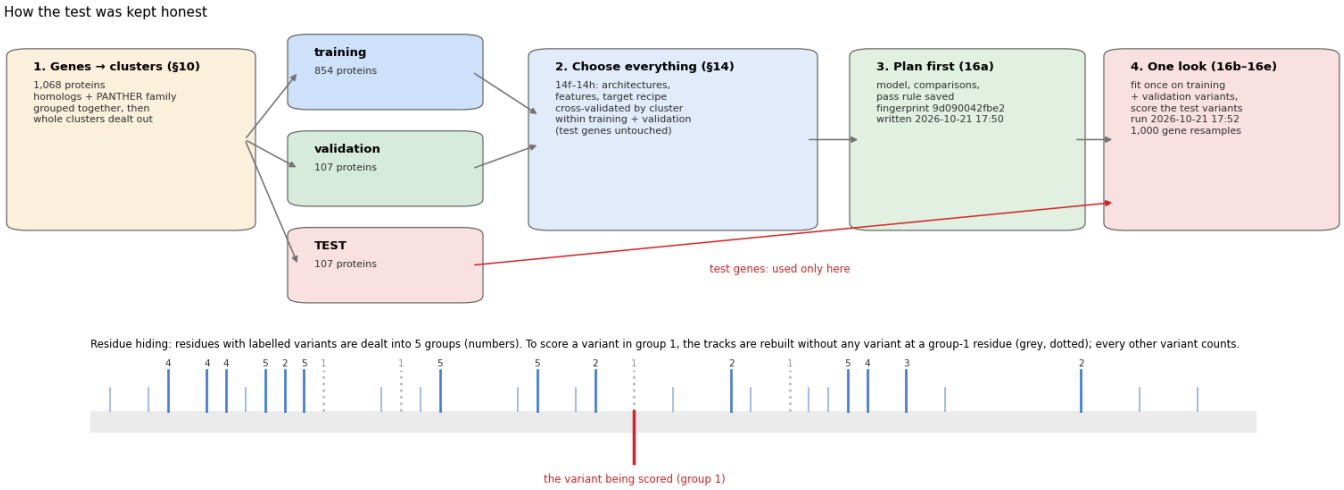


Figure 24. The safeguards that keep the test honest (from 16g): whole gene families in one split; every choice made without the test genes; the plan saved before the test; one run; uncertainty from resampling whole genes. Bottom: residue hiding.

The plan (16a) fixed the primary model (logistic regression on all 42 features, trained on all 15,339 training and validation variants), the comparisons, and four questions, each passing only if the whole 95% range of the difference is above 0:

Question	Result	95% range	Verdict
Q1: the model beats ESM-1v on all variants (0.890 vs 0.620)	+0.270	+0.048 to +0.447	**pass**
Q2: the model beats ESM-1v within genes (0.904 vs 0.749; 8 genes)	+0.156	−0.001 to +0.321	not shown
Q3: the model beats ESM-1v where ESM is unsure (0.879 vs 0.545)	+0.335	+0.182 to +0.474	**pass**
Q4: ESM + tolerance beats ESM alone (0.826 vs 0.620)	+0.206	+0.021 to +0.299	**pass**

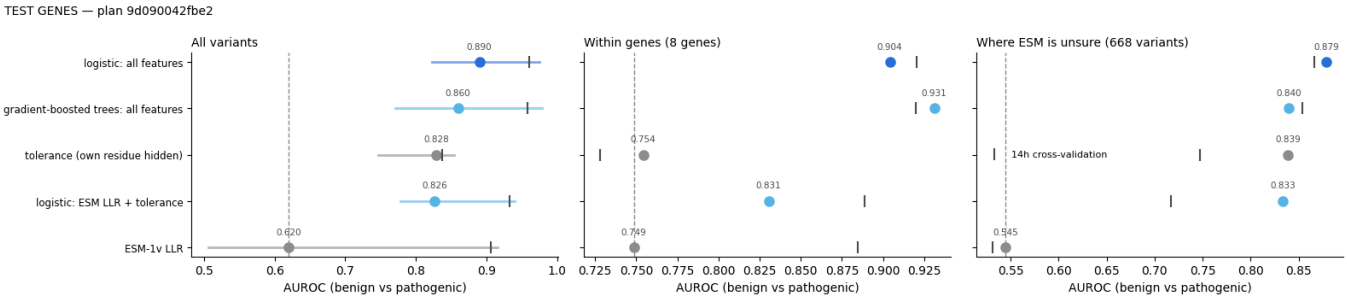


Figure 25. Every model on the test genes (from 16d). Small vertical ticks mark each model's 14h cross-validation score for comparison.

Model (§16 test genes, 1,357 variants)	All variants (95% range)	Within genes (8)	Where ESM is unsure (668)
ESM-1v LLR alone	0.620 (0.51–0.92)	0.749	0.545
Tolerance alone, own residue hidden	0.828 (0.75–0.85)	0.754	0.839
Logistic regression: ESM LLR + tolerance	0.826 (0.78–0.94)	0.831	0.833
Logistic regression: all 42 features (primary)	**0.890 (0.82–0.97)**	**0.904**	**0.879**
Gradient-boosted trees: all features	0.860 (0.77–0.98)	0.931	0.840

The 95% ranges are wide because three genes hold half of the test variants: a resample that leaves out RYR1, or includes it twice, moves the result a lot. That is why each question in the table above compares two models on the *same* resamples.

What the model relies on

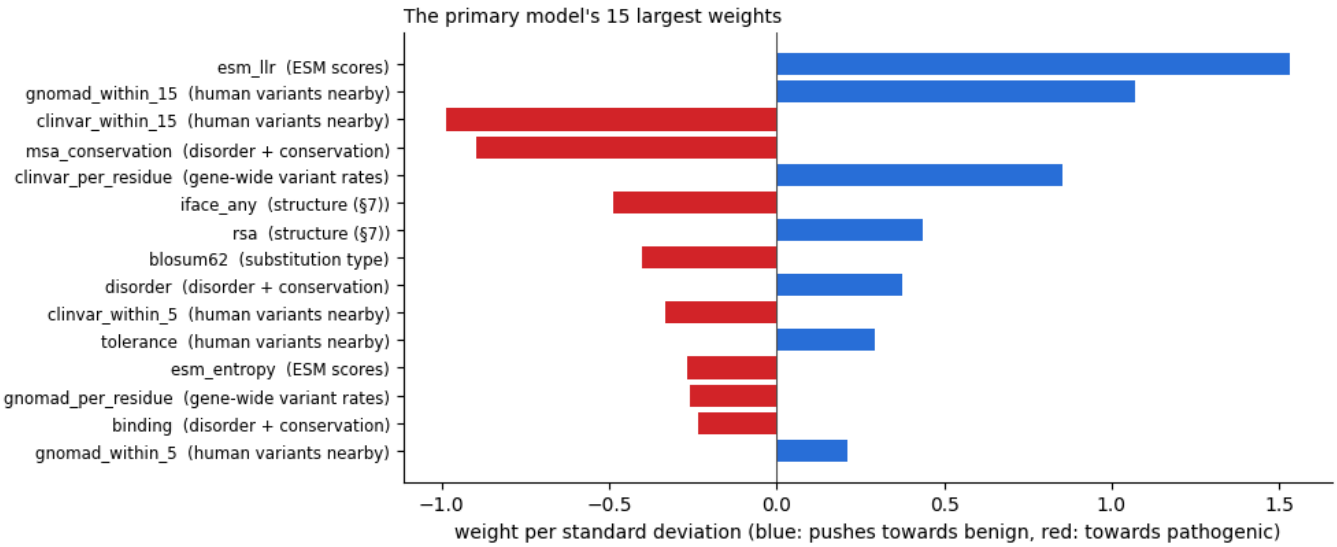


Figure 26. The final model's 15 largest weights (from 16c), per standard deviation of each feature. Blue pushes towards benign, red towards pathogenic. The strongest is ESM's LLR, followed by the number of gnomAD variants nearby (tolerated region) and the number of ClinVar variants nearby (disease hotspot).

Weights of features that move together (such as the ClinVar counts and the gene-wide ClinVar rate) can't be read one at a time: the model can trade them against each other. Read the top of the list as "what the model leans on", not as separate causes.

A closer look (descriptive, after the test)

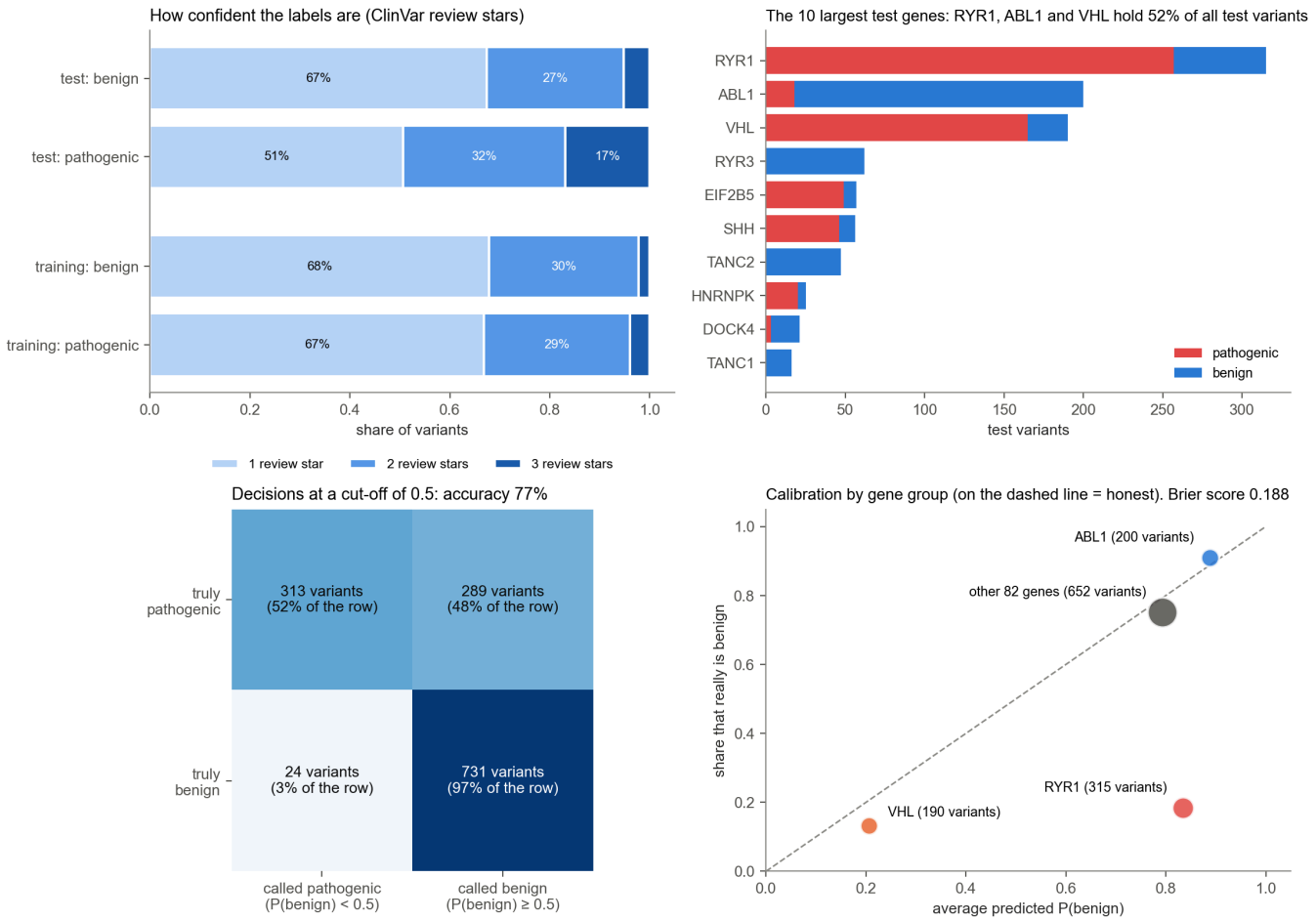


Figure 27. Labels and decisions on the test genes (redrawn from 16g). Top left: most labels in both sets rest on a single submitter (1 star); the test set's 3-star labels come almost entirely from RYR1's expert panel. Top right: three genes hold 52% of the test variants. Bottom left: at a cut-off of 0.5, 97% of benign variants are cleared but only 52% of pathogenic variants are caught. Bottom right: calibration by gene group. RYR1's variants get an average P(benign) of about 0.8, although most are pathogenic.

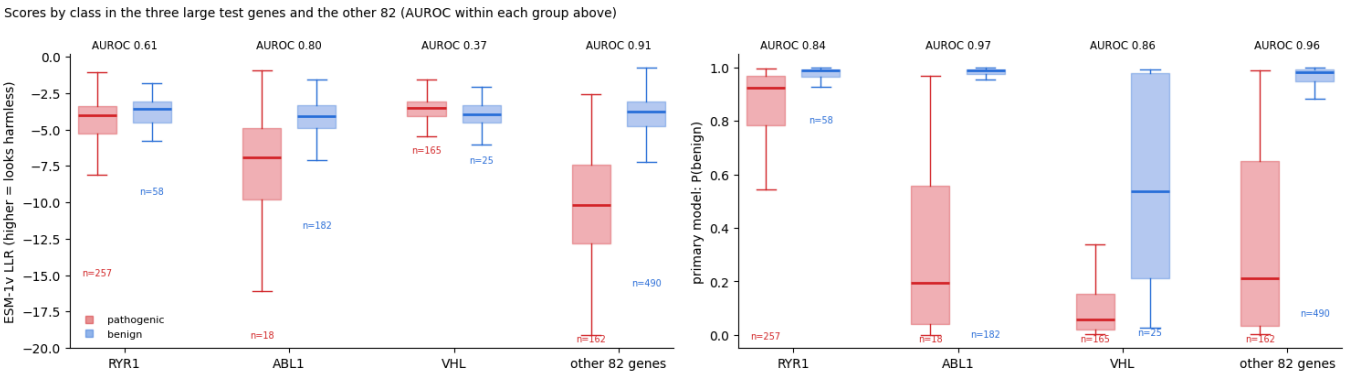


Figure 28. Why ESM-1v scored lower on the test genes (from 16g). In RYR1 and VHL, ESM gives pathogenic variants mild scores, close to ABL1's benign ones. In the other 82 test genes ESM-1v scores 0.911 and the model 0.963, the same gap as in 14h.

WHAT THE TEST SHOWS

- The model works on genes nobody chose (Q1), and helps most where evolution alone is at chance (Q3). Human variant data add to evolution in new genes (Q4).
- The size of the Q1 gap partly reflects this test set. Three genes hold half of it, and ESM struggles in two of them. In typical genes the gain is about +0.05 AUROC.
- The ranking is good, but the probabilities aren't yet trustworthy for new genes. Read them as ranks or recalibrate them. A plausible reason for the RYR1 miss: its expert-classified variants mostly concern malignant hyperthermia, which causes no symptoms without anaesthetic triggers, so carriers can look "healthy" in gnomAD.

9. What improved, and the way back to your original goal

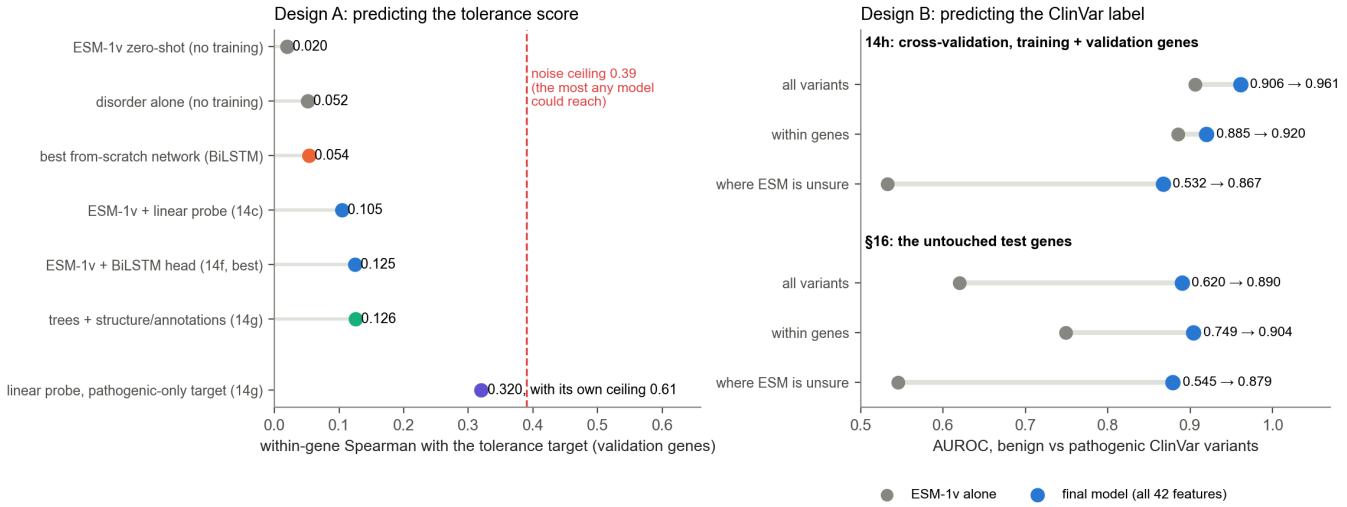


Figure 29. Every improvement in one place. Left (Design A): within-gene Spearman with the tolerance target on the validation genes, against the noise ceiling. Right (Design B): AUROC for ESM-1v alone (grey) and the final model (blue), in cross-validation and on the untouched test genes.

Metric	Baseline	Best result	Where
Within-gene Spearman with the tolerance target	0.020 (ESM-1v, no training)	0.125 (ESM + BiLSTM head)	14f, validation genes; ceiling 0.39
Same, with the pathogenic-only target	—	0.320 (linear probe)	14g; ceiling 0.61
AUROC, all variants	0.906 / 0.620	0.961 / 0.890	14h cross-validation / §16 test
AUROC within genes	0.885 / 0.749	0.920 / 0.904	14h / §16
AUROC where ESM is unsure	0.532 / 0.545	0.867 / 0.879	14h / §16
Accuracy at P(benign) ≥ 0.5	—	77%	§16 (52% of pathogenic caught)

Getting back to your original goal

Your goal was a model whose *prediction* is the tolerance score. Three routes lead there, and they can be combined:

1. **Train Design A on the better target.** Rebuild the target with the ClinVar side from pathogenic variants only (`CLINVAR_GROUPS = ("pathogenic",)` in §3), retrain the ESM heads on this laptop, then optionally run the LoRA fine-tuning on Biowulf (§15). The result is a sequence-only tolerance predictor with three times more learnable signal than now. It would need its own pre-registered test on the test genes. That is legitimate, because no Design A choice was based on the §16 result, but it should be reported as a separate analysis.
2. **Turn Design B into a tolerance score.** The final model can score *all 19 possible substitutions* at every residue: ESM, structure, annotations and nearby human variants exist for any residue. Averaging the 19 values of P(benign) gives one **model-predicted tolerance** per residue. It is available everywhere, including

residues where no variant has been seen, and it comes from a model whose variant-level accuracy has been checked on untouched genes. This is the quickest way to "a prediction score that determines tolerance". The per-residue average itself would still need its own check, for example against the observed track.

3. **Combine them.** Feed Design A's predicted tolerance into the variant model as one more feature (§16h), so sequence-learned tolerance and observed human variation inform each other.

IN PLAIN WORDS

Your original idea, predicting tolerance, is still alive. It was held back by a noisy target, not by the model. With the pathogenic-only target it becomes three times more learnable. Meanwhile, the variant model can already be turned into a per-residue tolerance score, built from a model whose accuracy has been checked on genes nobody chose.

10. Glossary

Term	Meaning
AUROC	the chance that a random benign variant gets a more benign score than a random pathogenic one. 0.5 = chance, 1 = perfect
BLOSUM62, Grantham	tables of how different two amino acids are (evolutionary substitution rates; chemical distance)
Bootstrap, 95% range	resampling the genes with replacement many times and keeping the middle 95% of the recomputed results
Calibration, Brier score	whether stated probabilities match observed frequencies; the Brier score is their mean squared error (0 = perfect)
Centring	subtracting each gene's median tolerance, so each track is measured from its own typical level
Cluster	a group of related genes (similar sequence or same PANTHER family), always kept in the same split
Cross-validation	fitting on 4/5 of the data and scoring the other 1/5, rotating, so every item is scored by a model that never saw it
Early stopping	ending training when the score on set-aside data stops improving
Epoch	one pass over all the training data
ESM-1v, ESM-2	protein language models (650 million weights) trained on natural protein sequences
Feature, input	a number the model is allowed to look at
Fine-tuning, LoRA	continuing to train a pre-trained model on a new task; LoRA trains small add-on matrices instead of all its weights
Frozen	a pre-trained model whose weights are kept fixed while something on top of it is trained
LLR	log-likelihood ratio: log P(new amino acid) − log P(original), from ESM. Negative = unexpected swap
Logistic regression	a weighted sum of features passed through an S-shaped curve to give a probability
Loss	a number measuring how far the model's output is from the target; training makes it smaller
Noise ceiling	the highest correlation any model could reach with a noisy target
Overfitting	memorising the training data instead of learning rules that carry over to new data
pLDDT	AlphaFold's confidence in its structure model at a residue (0–100)
Representation	ESM's internal description of a residue: 1,280 numbers summarising its context
Residue hiding	building a variant's human-variant features without any variant at its own residue
RSA	relative solvent accessibility: the share of a residue's surface exposed to water
Spearman correlation	agreement between two rankings: +1 same order, 0 unrelated, −1 reversed
Target	the answer a model learns to produce
Validation / test set	genes held out from training: validation for choosing between models, test for one final check
VUS	variant of uncertain significance: a ClinVar entry with no pathogenic or benign call